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Lesson #4.00
2365 - Level 3 City and Guilds

People think that electricity gives light. However, this is wrong. In modern lighting systems such as an LED which has a chemical inside. Once you provide the chemical with additional electrons it becomes hot. The heat creates the light.

DC is better for heating than AC.

Trains run on DC.

DC there is only resistance in ohms (RΩ)
Voltage drops happens because of the distance.

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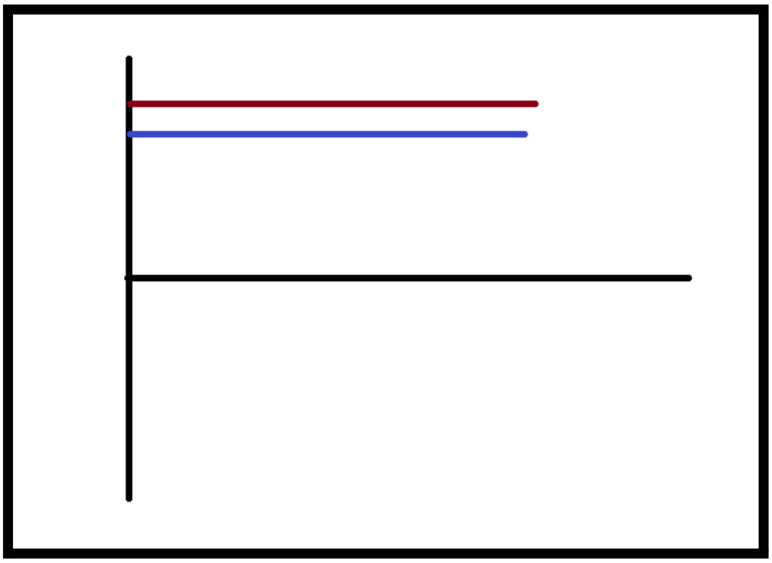
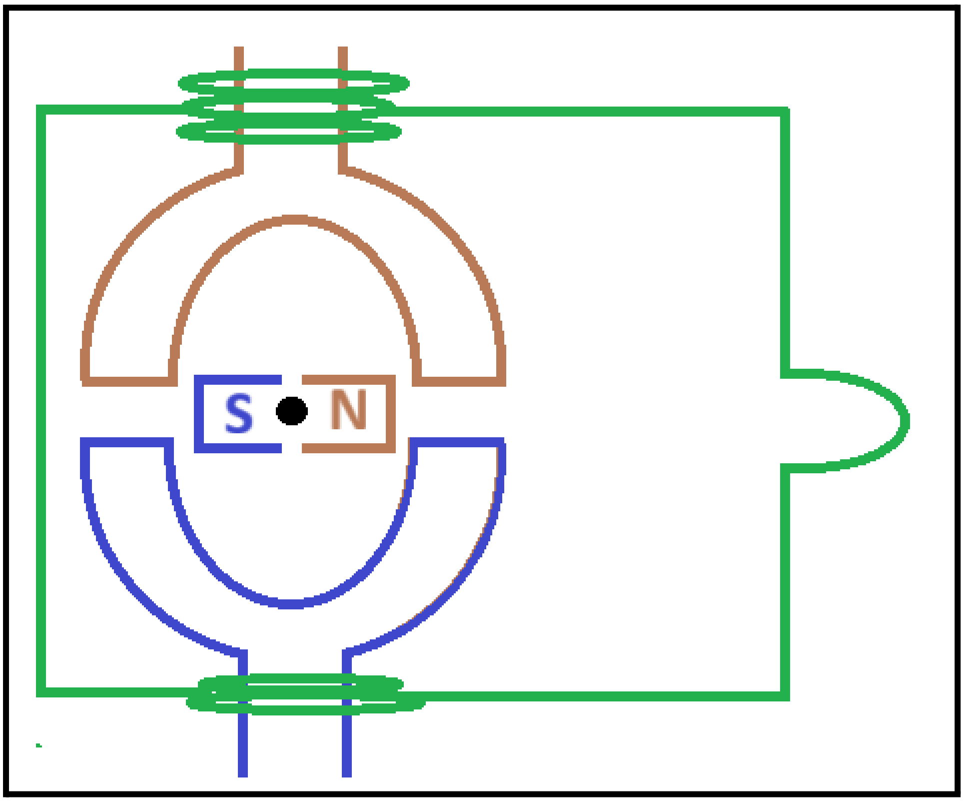


In phase, DC when you put pressure behind it flows.

An example, of DC is solar panel and battery.

In order to create electricity you must spin a magnet inside a magnet.



Electrons only flow in a complete circuit.
Because, electrons need to take the place of other moving electrons.
Electricity is about capturing magnetism.

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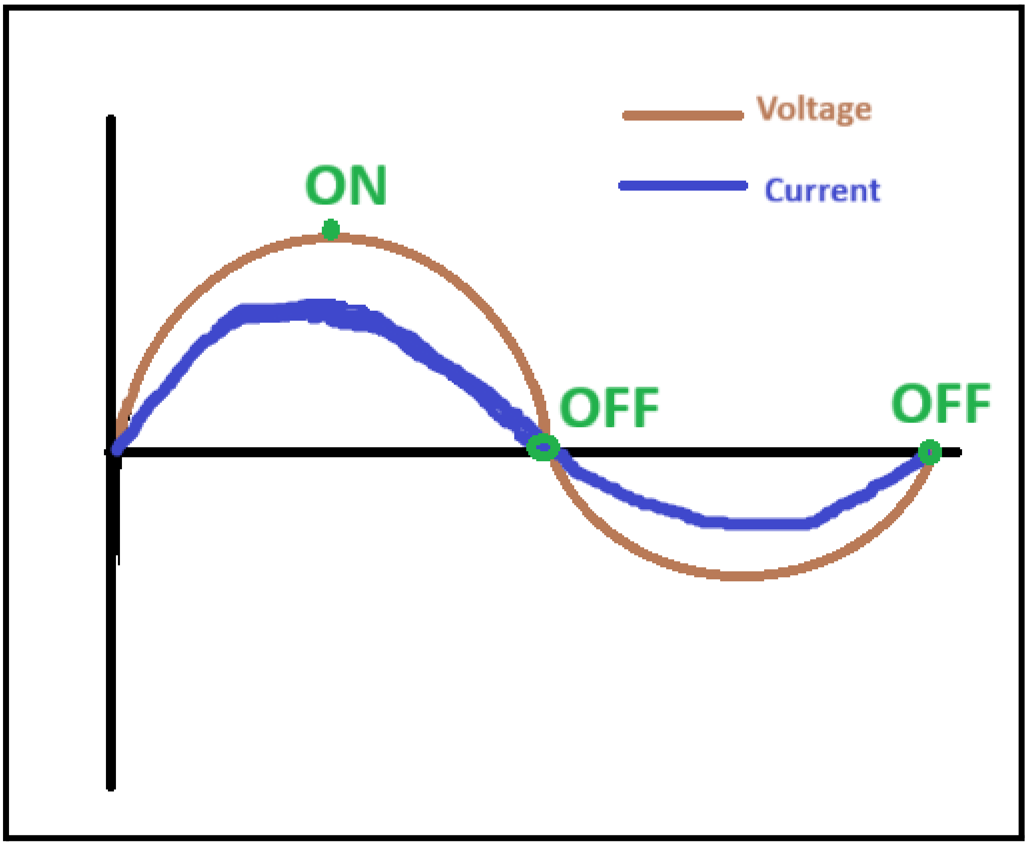

In the **OFF state** the magnet is completely flat. No electrical current is being induced.

Frequence is measured in Hertz.
Power = Current x Voltage. Measured in watts.

In phase/in sync current and voltage are in a purely-resistive load.

The voltage of the sockets is 230 volts.

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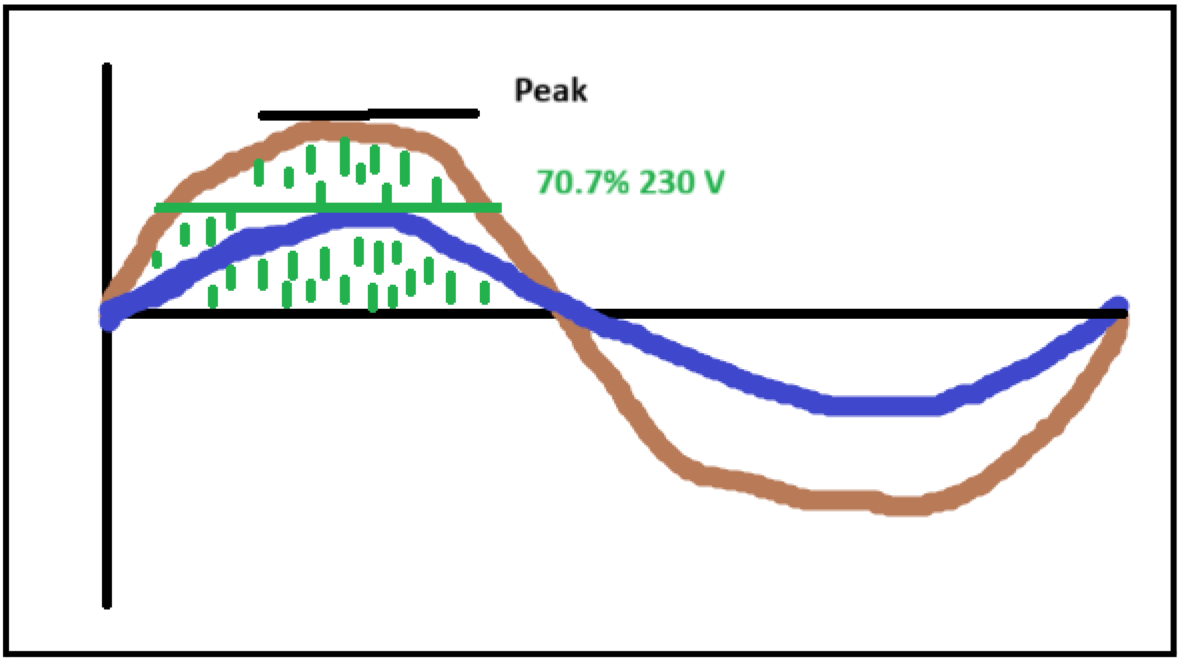

RMS => Root Mean Square

RMS is 0.707 which means 70%.

The Voltage is only at the peak for a very short time. So they wanted to find the DC equivalent which is 230 volts. This is the 70.7%.

Coil same as an inductor. Because, it induces a magnetic field.

When you put electricity through a coil it induces a magnetic field. When the magnetic changes position the magnetic field collapses and starts up again. Which is why the graph of an AC is in a wave like motion.

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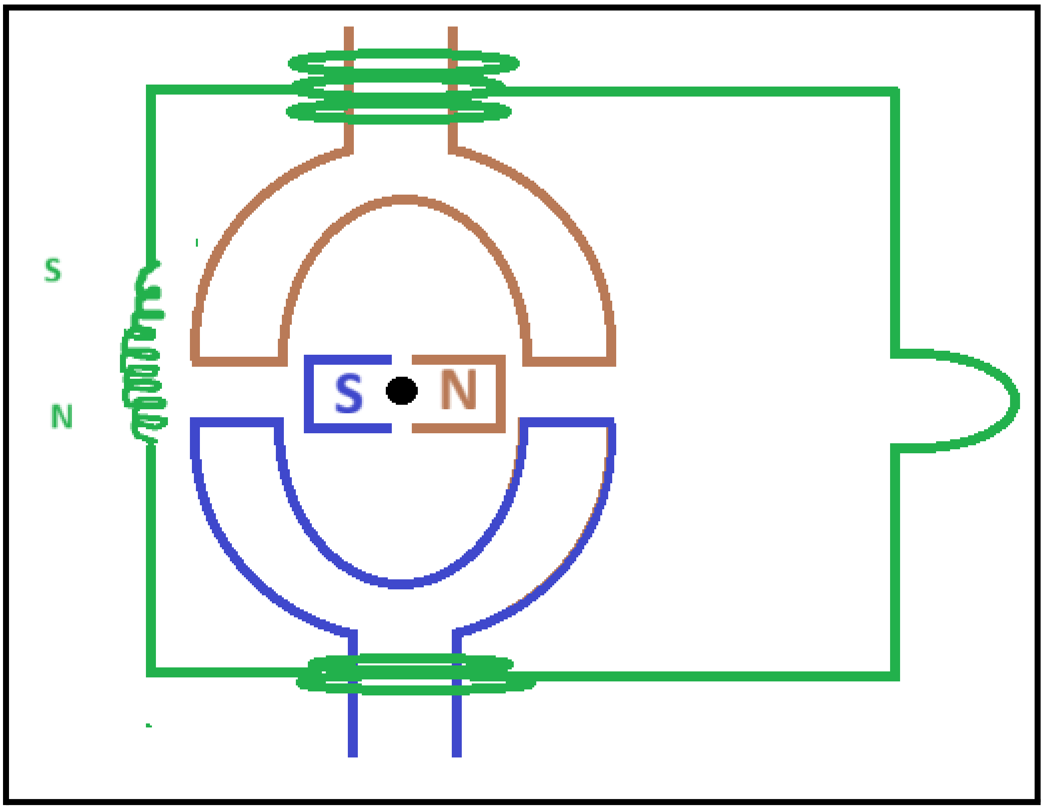

In an inductor you apply pressure (voltage) then there is movement (current).
Current lags voltage in an **inductor by 90 degrees**. Because, there is a moment where there is a plain state. Which causes the electrons to collapse.

The Breakdown of the resistances

XLΩ - Inductive reactance is measured in ohms Ω. This is a resistance in a coil (the inductor).

RΩ - We have resistance from the copper wire itself and the distance.

Z = total resistance which is **XLΩ + RΩ .**

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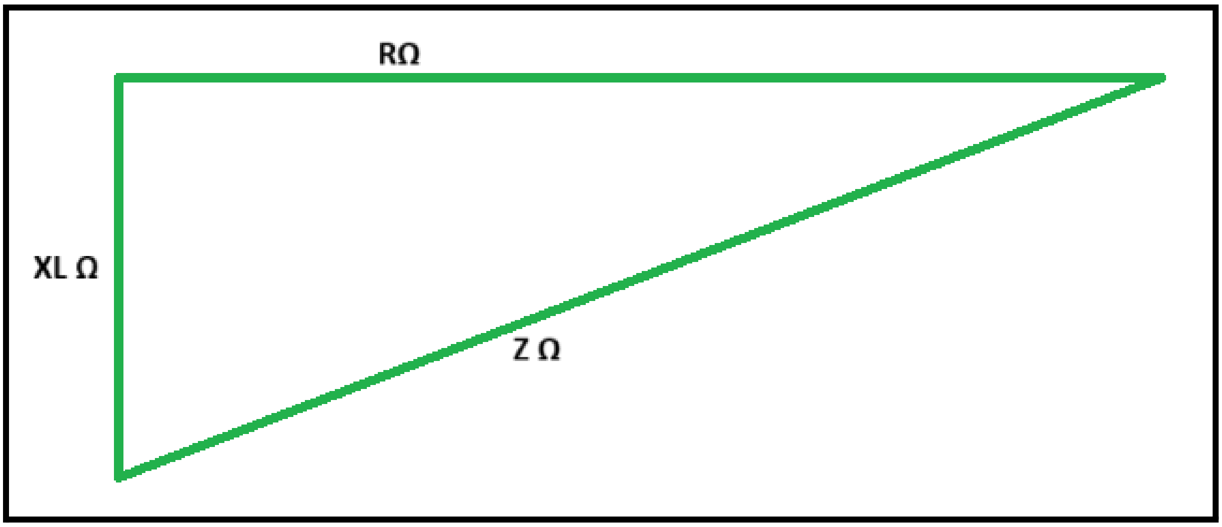

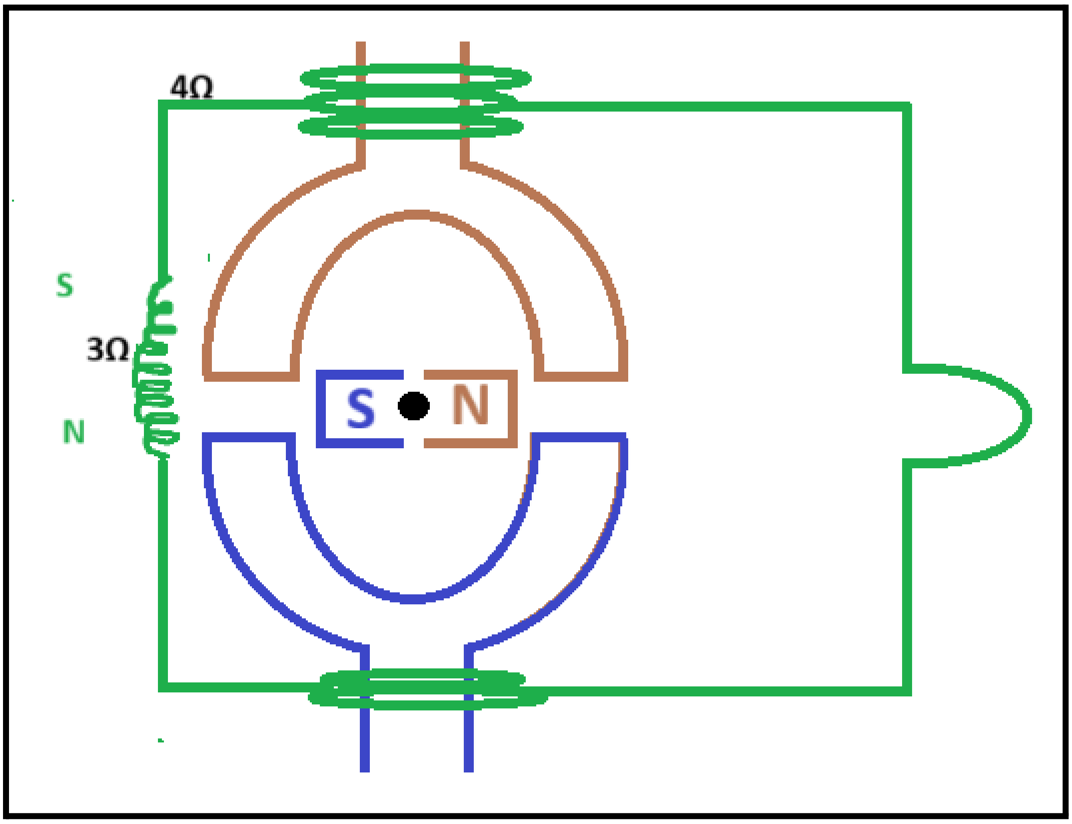

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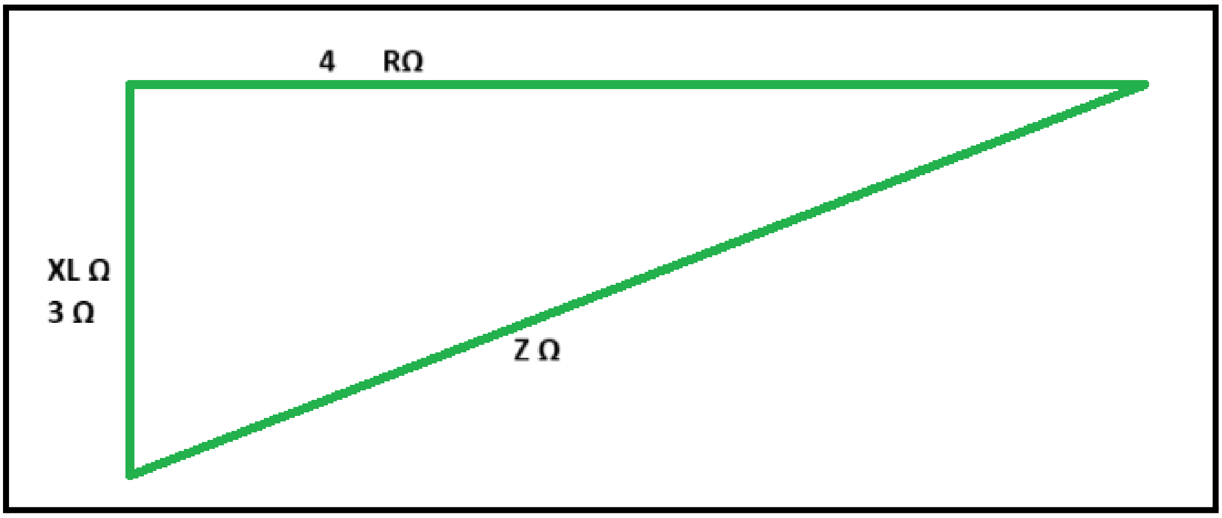
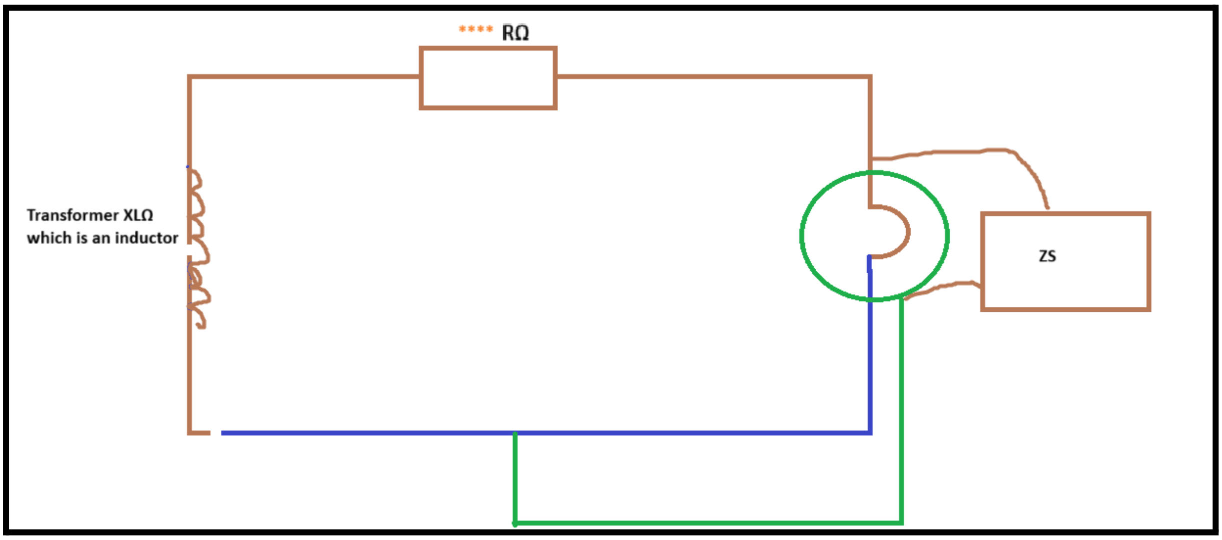


Image AA



Ze = X + R.

Zs = Ze + (R1 + R2).

Ze is from the transformer to **Image AA** ****

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Part one

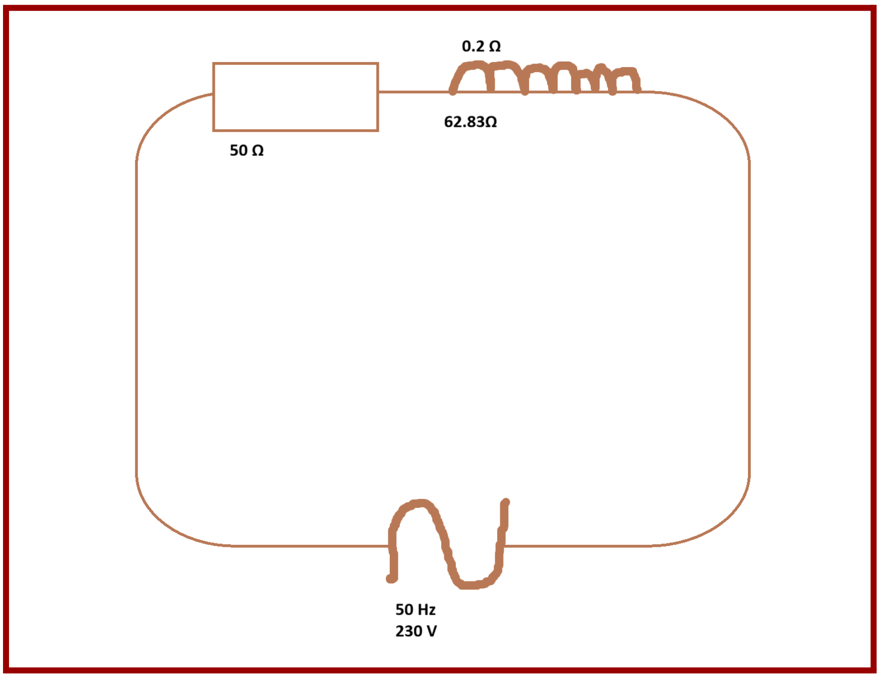
A coil will become magnetised. The number of coils is measured in Henrys.
This calculation is to determine how much resistance is in the coil. The resistance in a coil is stored in a variable called **XL**.

XL = **2πFL**

XL = **2 X π X F X L** (**Frequency**, **Henry**)

2 X 3.14 X 50 Hz X 0.2

Answer: 02.83Ω



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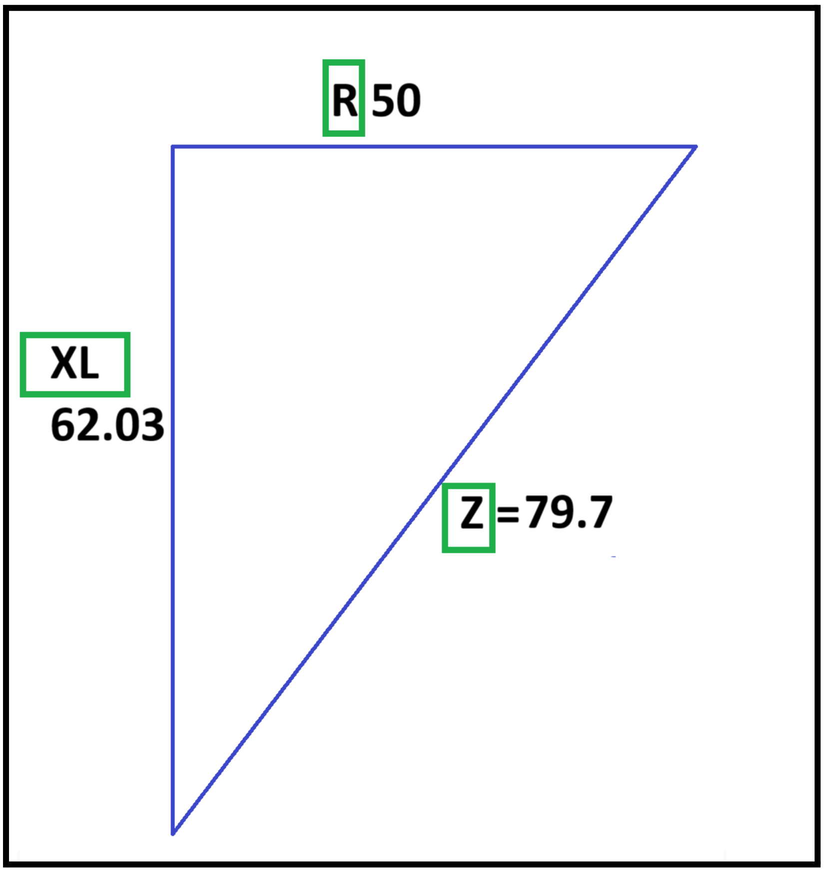

$$\sqrt{62.03^2 + 50^2} \approx \boxed{79.673}$$

Total resistance is stored into a variable called **Z**.

Z = **XL** + **R**.
Total resistance = Inductive resistance (coil resistance) + resistance (resistance in the copper itself)

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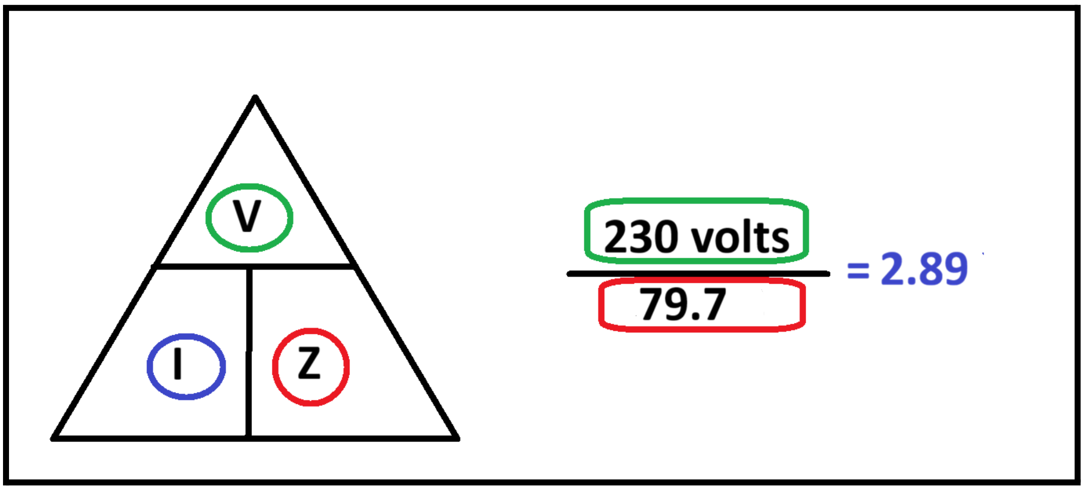

End of Part one

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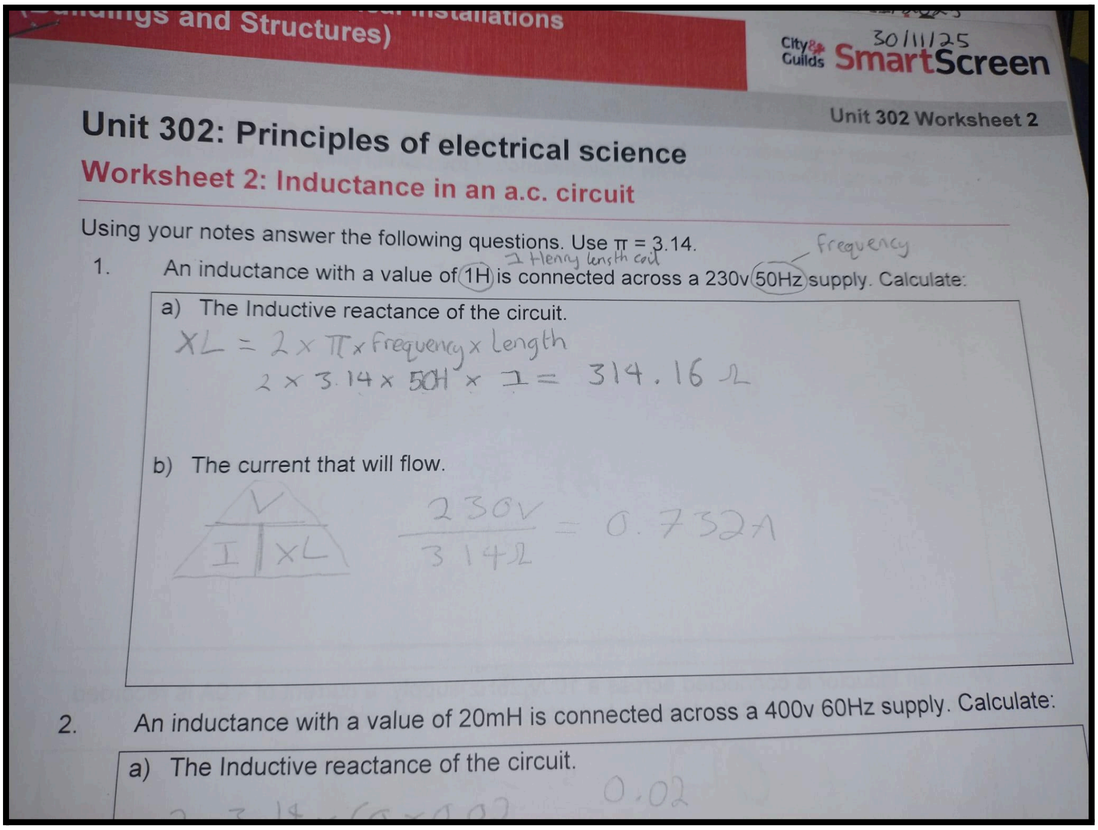
Worksheet 2 - WS2

WS2 Question 1



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Unit 302 Worksheet 2

Unit 302: Principles of electrical science

Worksheet 2: Inductance in an a.c. circuit

Using your notes answer the following questions. Use π = 3.14.

1. An inductance with a value of 1H is connected across a 230v 50Hz supply. Calculate:

1 Henry length coil — Frequency

a) The Inductive reactance of the circuit.

XL = 2 x π x frequency x length

2 x 3.14 x 50H x 1 = 314.16 Ω

b) The current that will flow.

V / I XL

230V / 314Ω = 0.732A

2. An inductance with a value of 20mH is connected across a 400v 60Hz supply. Calculate:

a) The Inductive reactance of the circuit.

0.02

Question 1 a) The inductive reactance of the circuit

XL = **2πFL**

XL = **2 X π X F X L** (**Frequency**, **Henry**)

2 X 3.14 X 50 H X 1 = 314.16 Ω

Question 1 b) The current that will flow:

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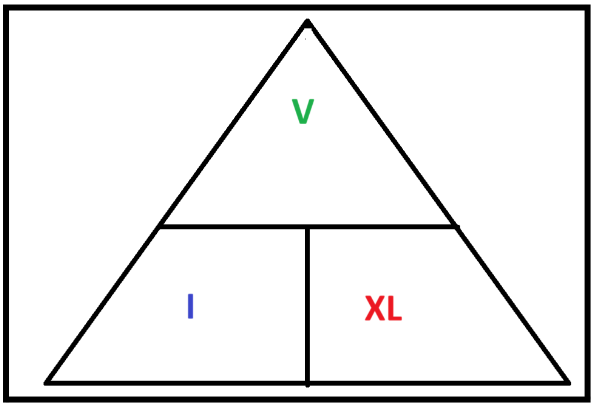
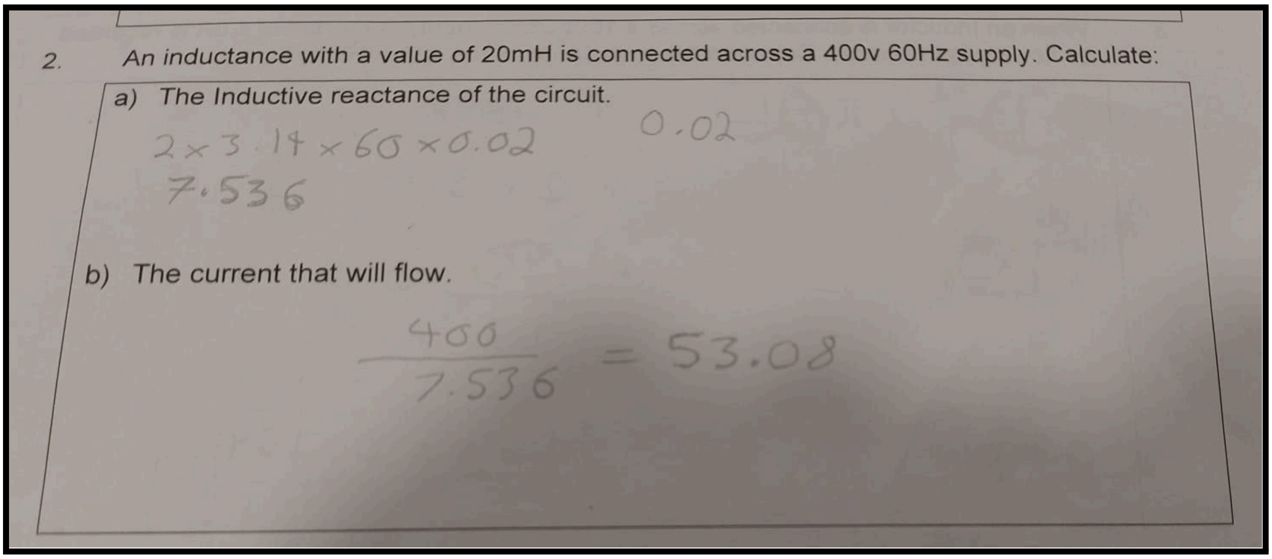


230 V / 314Ω = 0.732 amps

WS2 Question 2



2. An inductance with a value of 20mH is connected across a 400v 60Hz supply. Calculate:

a) The Inductive reactance of the circuit.

2 × 3.14 × 60 × 0.02 0.02
7.536

b) The current that will flow.

$\frac{400}{7.536} = 53.08$

Question 2 a) The Inductive reactance of the circuit.

Two pies for lunch

XL = $2\pi FL$

XL = 2 **X** π **X F X** L (**Frequency**, **Henry**)

2 X **3.14** X **60 Hz** X **20** (**mH**, **millihenry**)

= 7.536 (2 d.p)
Answer = 7.54 ohms

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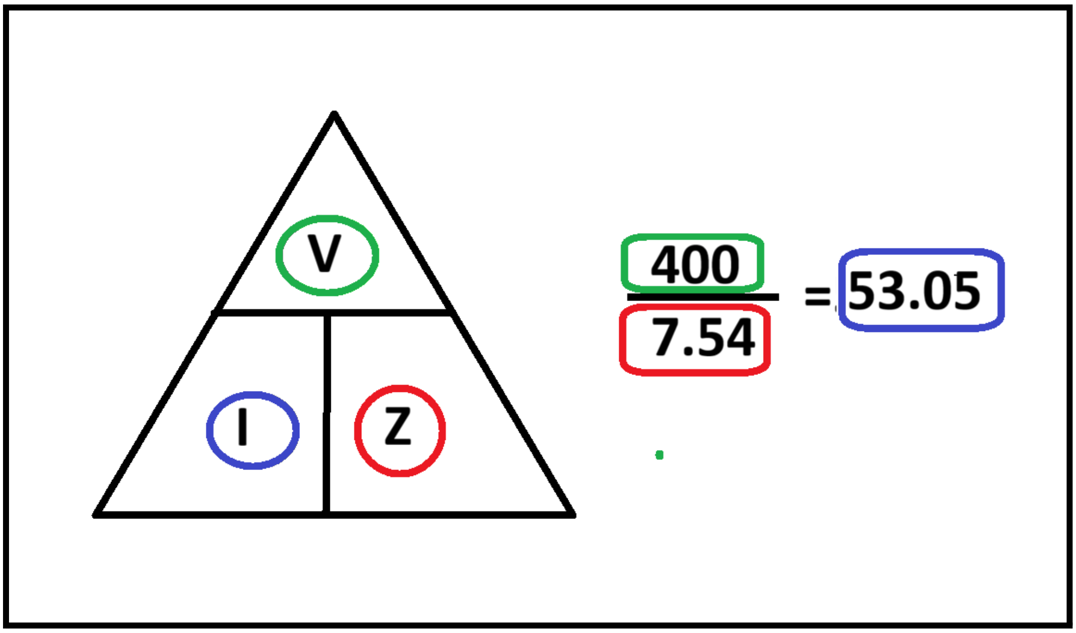
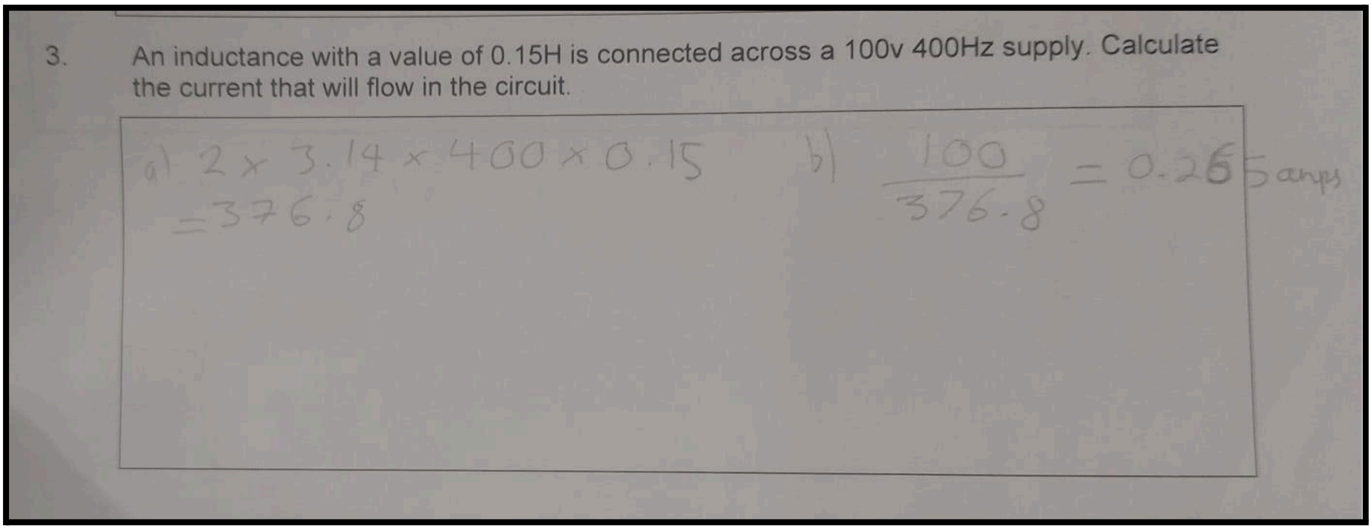
Question 2 b) The current that will flow



Current is 53.05 amps

WS2 Question 3

Question 3 An inductance with a value of 0.15H is connected across a 100v 400Hz supply. Calculate the current that will flow in the circuit.



3. An inductance with a value of 0.15H is connected across a 100v 400Hz supply. Calculate the current that will flow in the circuit.

a) $2 \times 3.14 \times 400 \times 0.15$
$= 376.8$

b) $\frac{100}{376.8} = 0.265$ amps

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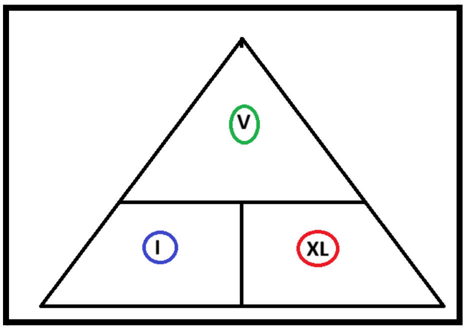
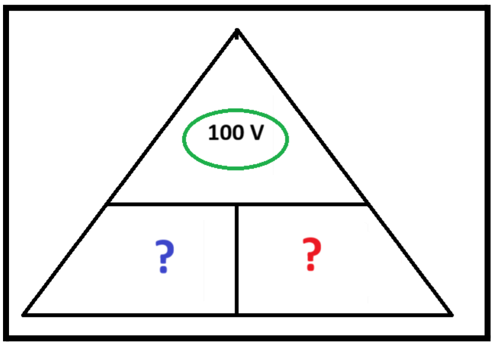
We know:

- **400 Hz**
- **0.15 Henrys**
- **100 V**





Step a)
Calculate the inductance XL first.

XL = **2 X π X F X L** (**Frequency Hz**, **Henry H or millihenry mH**)

2 X 3.14 X 400 Hz X 0.15 H

Inductance XL = 376.8 Ω

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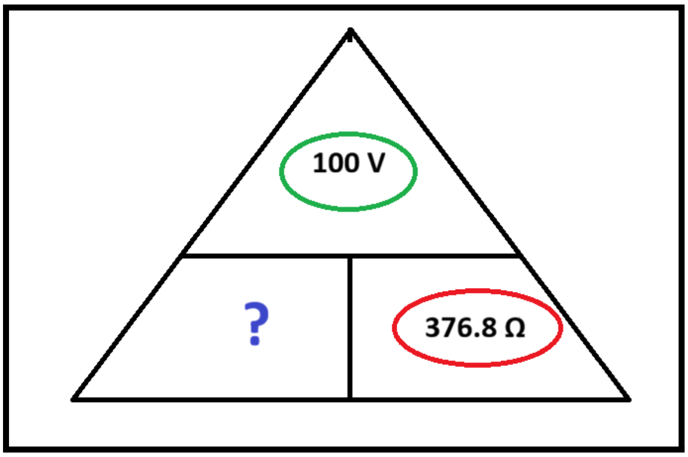
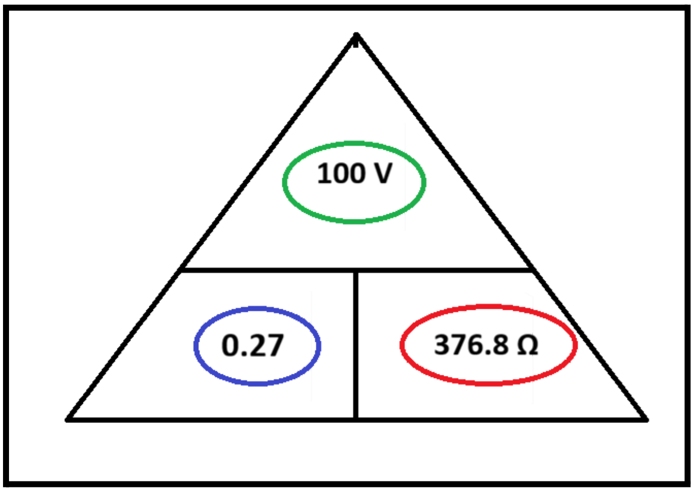


Step b) Calculate the current that will flow in the circuit.

100 V / 376.8Ω = 0.265 (2 d.p)

= 0.27 amps



WS2 Question 4

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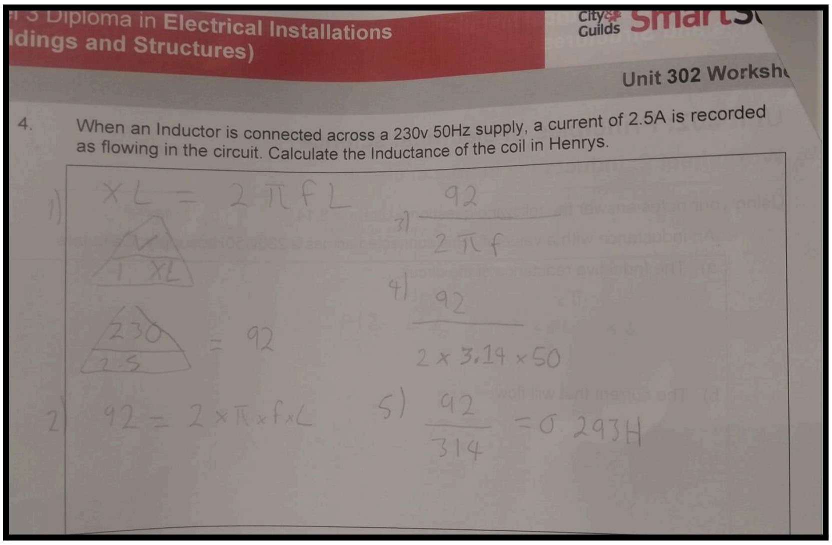


Question 4

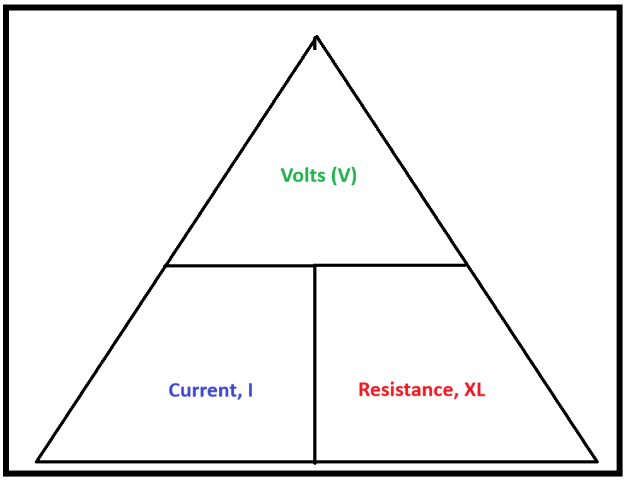
When an inductor is connected across a 230v 50 Hz supply, a current of 2.5A is recorded as flowing in the circuit. Calculating the inductance of the coil in Henrys.

What we know:

- 230 volts
- 50 Hertz Hz
- 2.5 Amps

Put the information into a triangle.

Step a)



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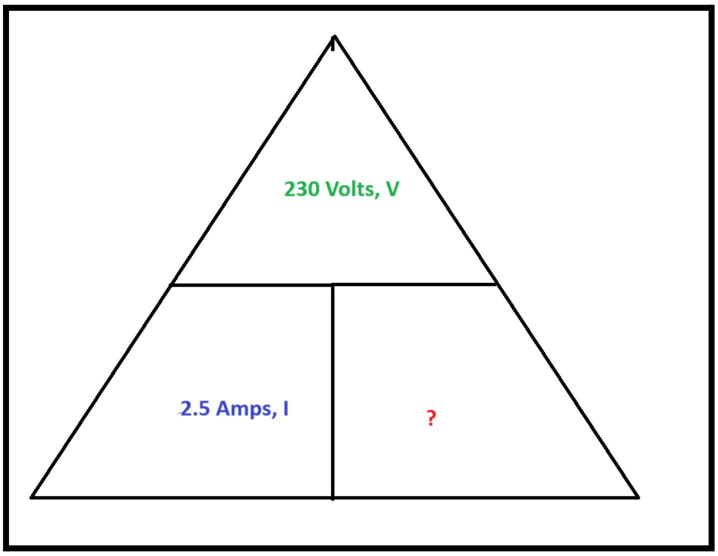
Step b)



XL = **230 volts** / **2.5 amps**

Inductive reactance = 92

Step c)

92 = **2 X π X F X L** (**Frequency Hz**, **Henry H or millihenry mH**)

92 = **2** X **3.14** X **50Hz** X **?**

$$\frac{92}{(2 \times 3.14 \times 50\,\mathrm{Hz} \times ?)}$$

Step d)

$$\frac{92}{314} = 0.293$$

Answer = 0.293 H (Henrys)

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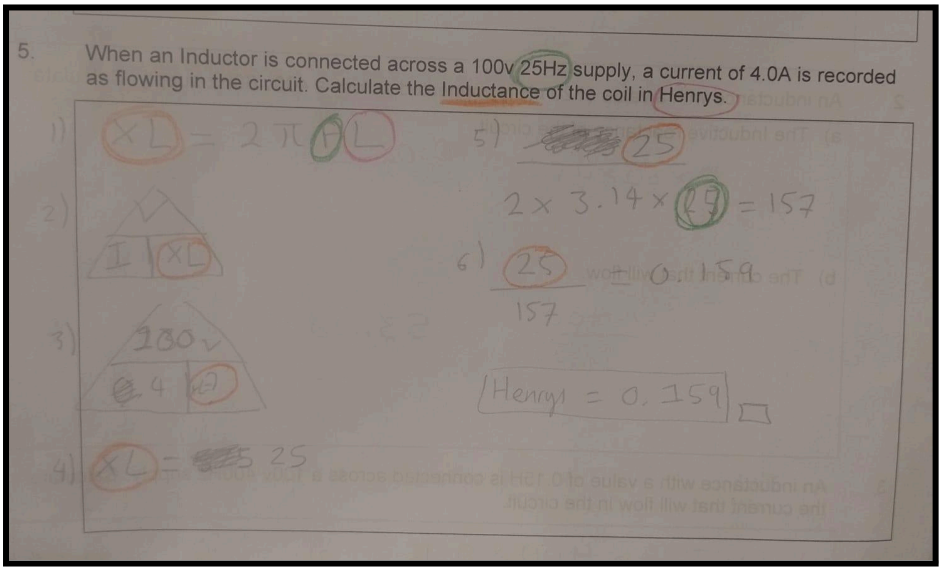
WS2 Question 5


5. When an Inductor is connected across a 100v 25Hz supply, a current of 4.0A is recorded as flowing in the circuit. Calculate the Inductance of the coil in Henrys.

1) XL = 2 π f L
2) V / I XL
3) 100v / 4 | (25)
4) XL = 25
5) 25 / 2 x 3.14 x 25 = 157
6) 25 / 157 = 0.159
Henrys = 0.159

When an Inductor is connected across a 100v, 25 Hz supply, a current of 4.0A is recorded as flowing in the circuit. Calculate the Inductance of the coil in Henrys?

Step one

What we know:

- **100 volts**
- 25 Hertz Hz
- **4.0 Amps**

XL = **2** **X** **π** **X** **F** **X** **L** (**Frequency Hz**, **Henry H or millihenry mH**)

XL = **2** **X** **3.14** **X** **25 Hz** **X** **?**

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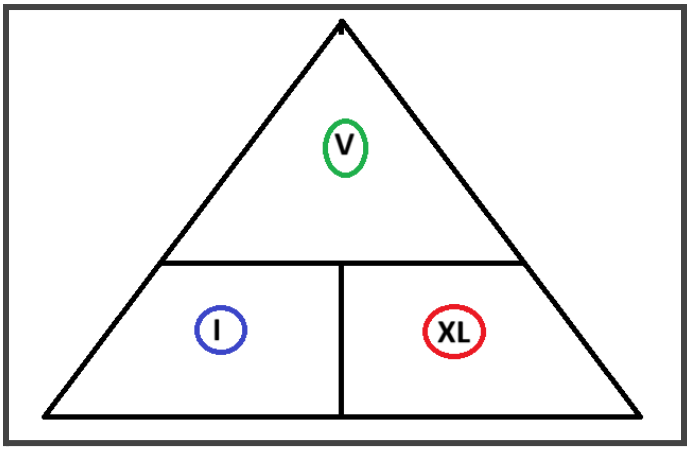
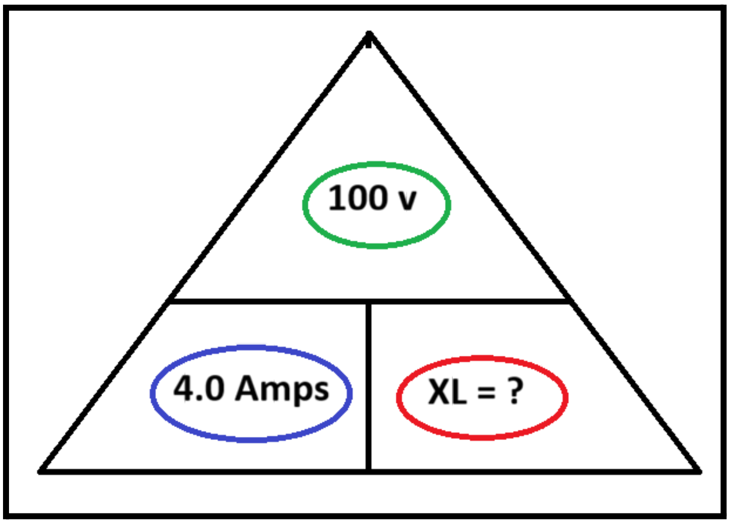
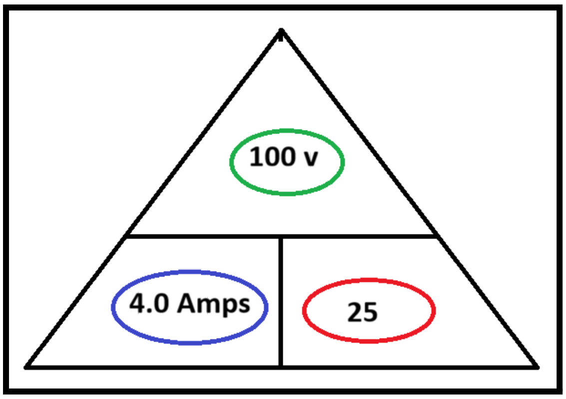


Step two



100 / 4.0 = 25

Step three



Step four

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$$\boxed{\frac{25\ (\text{Inductive Reactance } X_L)}{(2 \times 3.14 \times 25\ \text{Hz} \times ?\ \text{Henrys})}}$$

Step five

$$\boxed{\frac{25\ (\text{Inductive Reactance } X_L)}{157}}$$

Step six

$$\frac{25}{157} = \boxed{0.159\ \text{H}}$$

Great! To convert henrys (H) to millihenrys (mH), we multiply by 1,000:

$$0.159\,\text{H} \times 1000 = 159\,\text{mH}$$

✅ So:

$$\frac{25}{157} \approx 0.159\,\text{H} = 159\,\text{mH}$$

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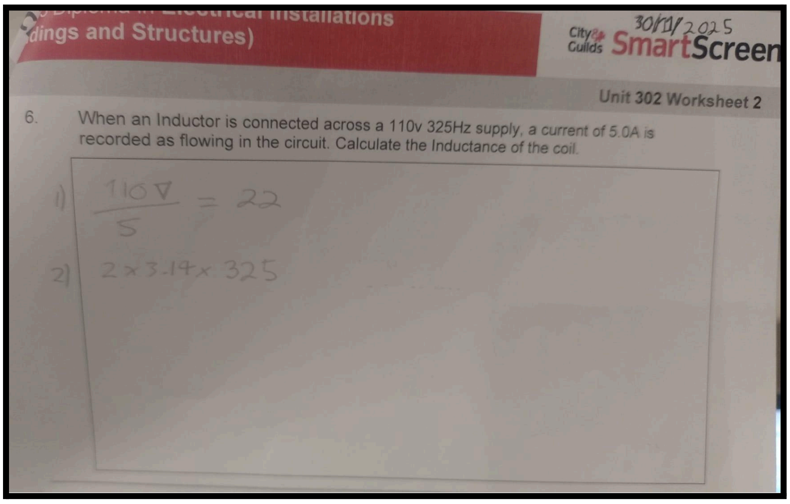
WS2 Question 6



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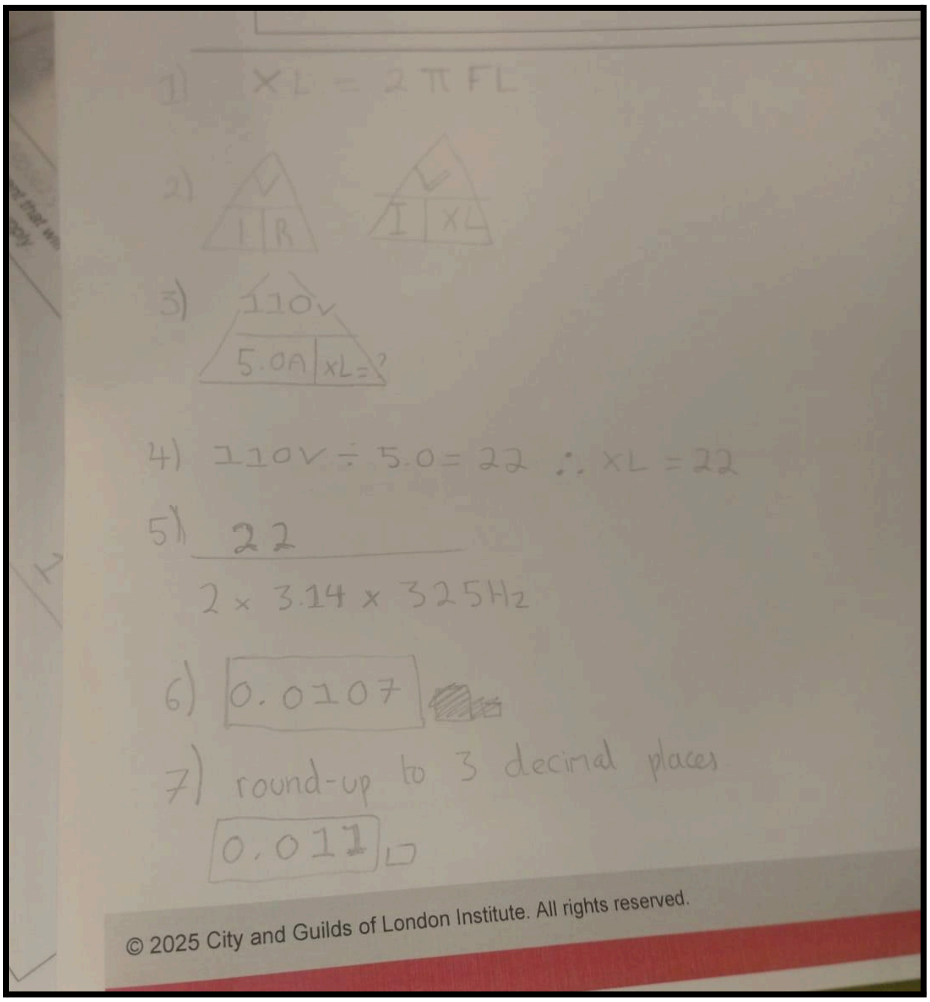
Unit 302 Worksheet 2

6. When an Inductor is connected across a 110v 325Hz supply, a current of 5.0A is recorded as flowing in the circuit. Calculate the Inductance of the coil.

1) $\frac{110V}{5} = 22$

2) $2 \times 3.14 \times 325$



1) $X_L = 2\pi F L$

2) V / I | R   V / I | XL

3) 110v / 5.0A | XL=?

4) $110V \div 5.0 = 22 \therefore X_L = 22$

5) $\frac{22}{2 \times 3.14 \times 325Hz}$

6) $\boxed{0.0107}$

7) round-up to 3 decimal places

$\boxed{0.011}$

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Question 6

When an Inductor is connected across a 110v 325Hz supply, a current of 5.0A is recorded as flowing in the circuit. Calculate the Inductance of the coil.

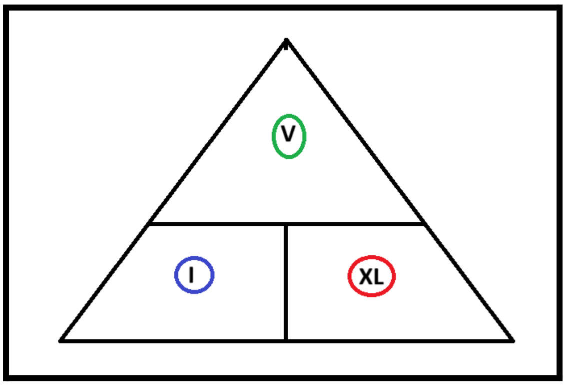
What we know:

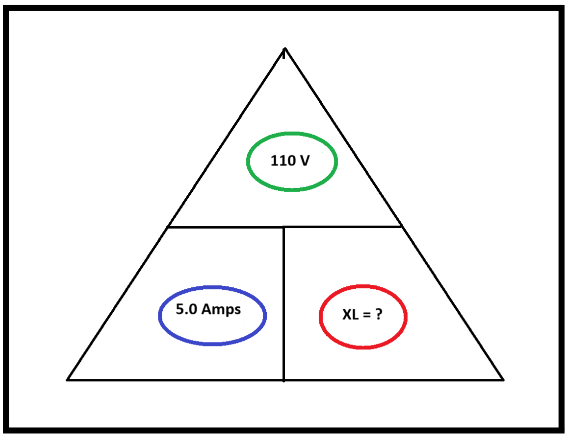
- 110 Volts
- 325 Hertz
- 5.0 amps
- ? Unknown is the Henrys.

XL = **2 X π X F X L** (**Frequency Hz**, **Henry H or millihenry mH**)

XL = **2 X 3.14 X 325 Hz X ?**

Step one





110 V / 5.0 = 22 Ω

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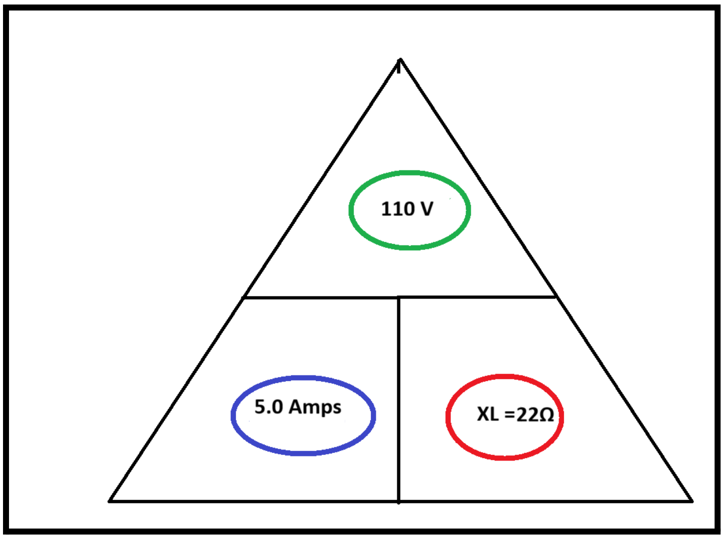

The answer is 22 Ω for the Inductive reactance.

Step two - (This is an additional step to calculate the Inductance of the coil in Henrys. However, it is not mandatory because the question states "Calculate the Inductance of the coil")

XL = **2 X π X F X L** (**Frequency Hz**, **Henry H or millihenry mH**)

22 Ω = **2 X 3.14 X 325 Hz X ?**

$$\frac{22\ \Omega}{2 \times 3.14 \times 325\ \text{Hz}} \approx 0.0108\ \text{Henrys}$$

= 0.0108 Henrys (2 D.P)
= 0.011 Henrys

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Worksheet 3 - WS3

WS3 - Question 1